

Topic 14: Domain-Specific Prompting

14.1 Prompting for Code

- Be precise: specify language, version, libraries, and any constraints
- Include context: existing code, function signatures, data structures the code must work with
- Ask for: type hints, docstrings, error handling, unit tests separately if needed
- For debugging: provide the code, the error message, and the expected vs actual behaviour
- Always review AI-generated code -- it can introduce subtle bugs or security issues

```
Write a Python 3.11 function using only the standard library.  
Function: parse_log_line(line: str) -> dict  
Input format: '2024-01-15 14:32:01 ERROR auth.py:52 Invalid token'  
Output: {date, time, level, file, line_number, message} as strings  
Requirements:  
- Use regex for parsing  
- Raise ValueError with a descriptive message if the format doesn't match  
- Include a docstring and 3 pytest test cases
```

14.2 Prompting for Data Analysis

- Provide the data schema or a sample of the data at the start
- State the analytical question precisely -- avoid ambiguous terms like 'analyse'
- Ask for SQL, Python (pandas), or a specific tool if you have a preference
- Request explanation of the logic alongside the code
- For insights: 'List the top 3 actionable insights from this data, supported by numbers'

```
You are a data analyst. I have a PostgreSQL table: sales(id, date, product, region, revenue)  
  
Write a SQL query that:  
1. Returns the top 5 products by total revenue in Q1 2024 (Jan-Mar)  
2. Shows: product name, total revenue, % of Q1 total revenue  
3. Orders by total revenue descending  
  
Then explain what each clause does in plain English.
```

14.3 Prompting for Creative Writing

- Provide: genre, tone, audience, length, and any must-include elements
- Give a style reference: 'Write in the style of [author/work]'

- Use constraints creatively: they focus the model and prevent generic output
- For long-form: generate an outline first, then expand section by section (prompt chaining)
- Iterate: generate 3 opening paragraphs, pick the best, then continue from there

```
Write the opening paragraph of a science fiction short story.  
Genre: Hard sci-fi / psychological thriller  
Setting: A research station on Europa (Jupiter's moon) in 2157  
Protagonist: A marine biologist who just discovered something impossible  
Tone: Tense, clinical, with a creeping sense of dread  
Length: 100-120 words  
Do NOT use the words 'darkness', 'silence', or 'suddenly'.
```

14.4 Prompting for Summarisation

- Specify: length, format (bullets vs prose), focus areas, and audience
 - Extractive summary: 'Quote the 3 most important sentences verbatim'
 - Abstractive summary: 'Rewrite the key ideas in your own words'
 - Focused summary: 'Summarise only the financial implications mentioned in this report'
 - Multi-document: 'Summarise these 3 articles and highlight where they agree and disagree'
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